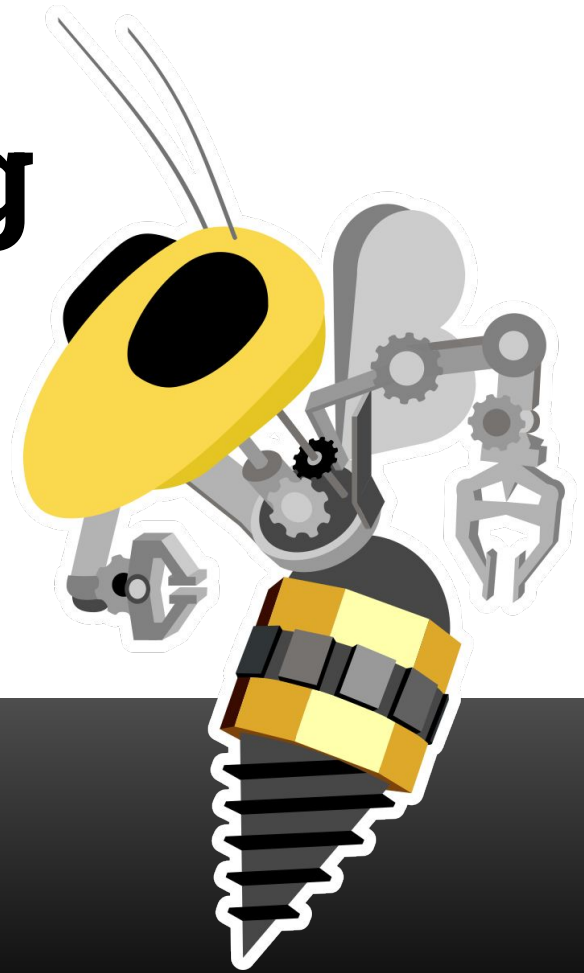


Firmware Training Week 5

State Machines + RTOSes

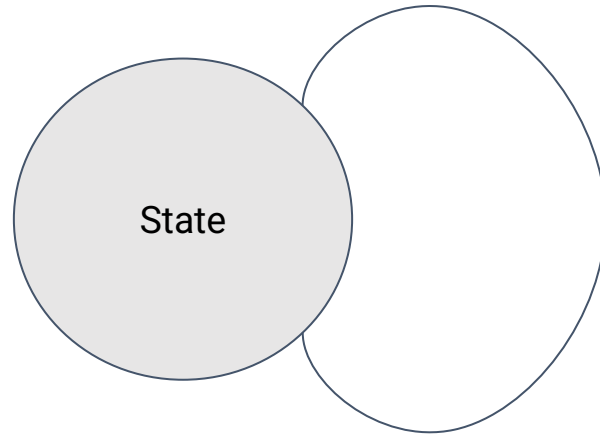


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State Machines

Transitions



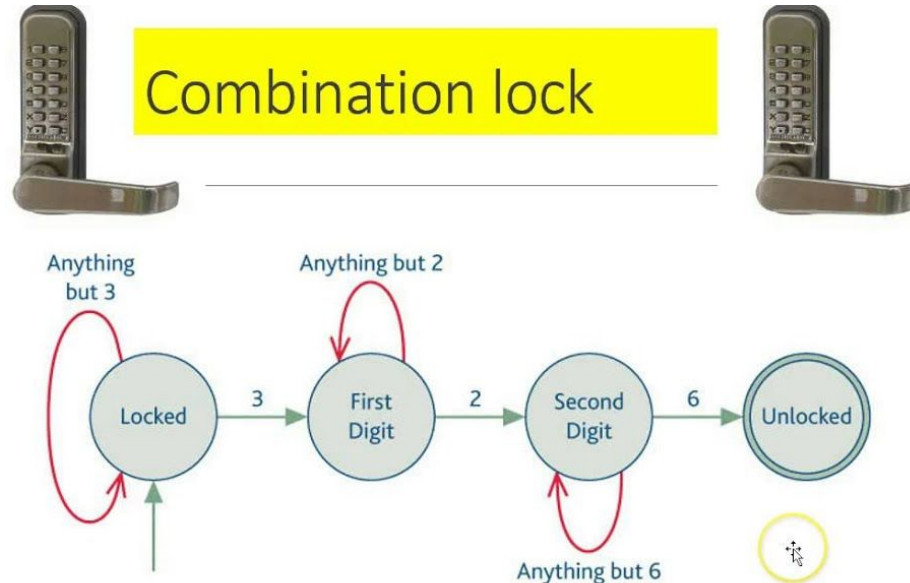
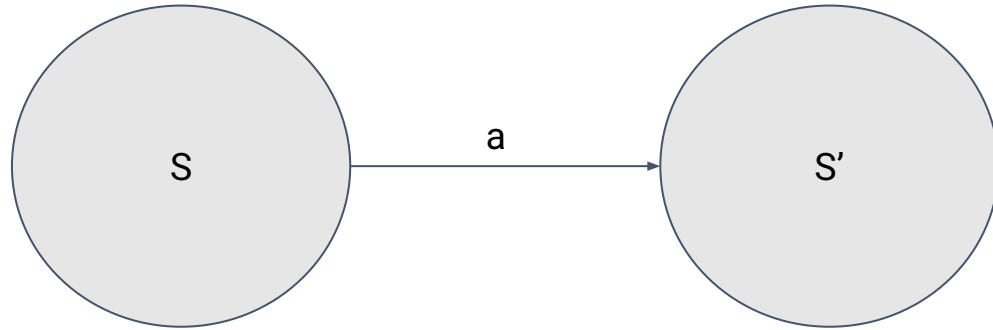
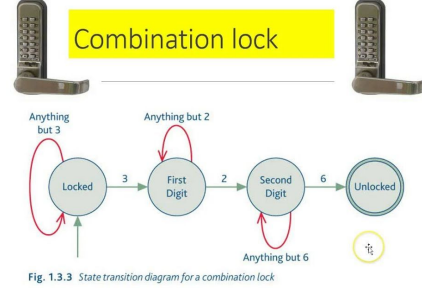


Fig. 1.3.3 State transition diagram for a combination lock





s	a	s'
Locked	Anything but 3	Locked
Locked	3	First Digit
First Digit	Anything but 2	First Digit
First Digit	2	Second Digit
Second Digit	Anything but 6	Second Digit
Second Digit	6	Unlocked

```
enum State {
    Locked,
    FirstDigit,
    SecondDigit,
    Unlocked
};

uint8_t input = 0;
State state = State::Locked;

switch (state) {
    case State::Locked:
        if (input == 3) {
            state = State::FirstDigit;
        }
        break;
    case State::FirstDigit:
        if (input == 2) {
            state = State::SecondDigit;
        }
        break;
    case State::SecondDigit:
        if (input == 6) {
            state = State::Unlocked;
        }
        break;
    case State::Unlocked:
        break;
}
```

Real Time Operating Systems (RTOS)

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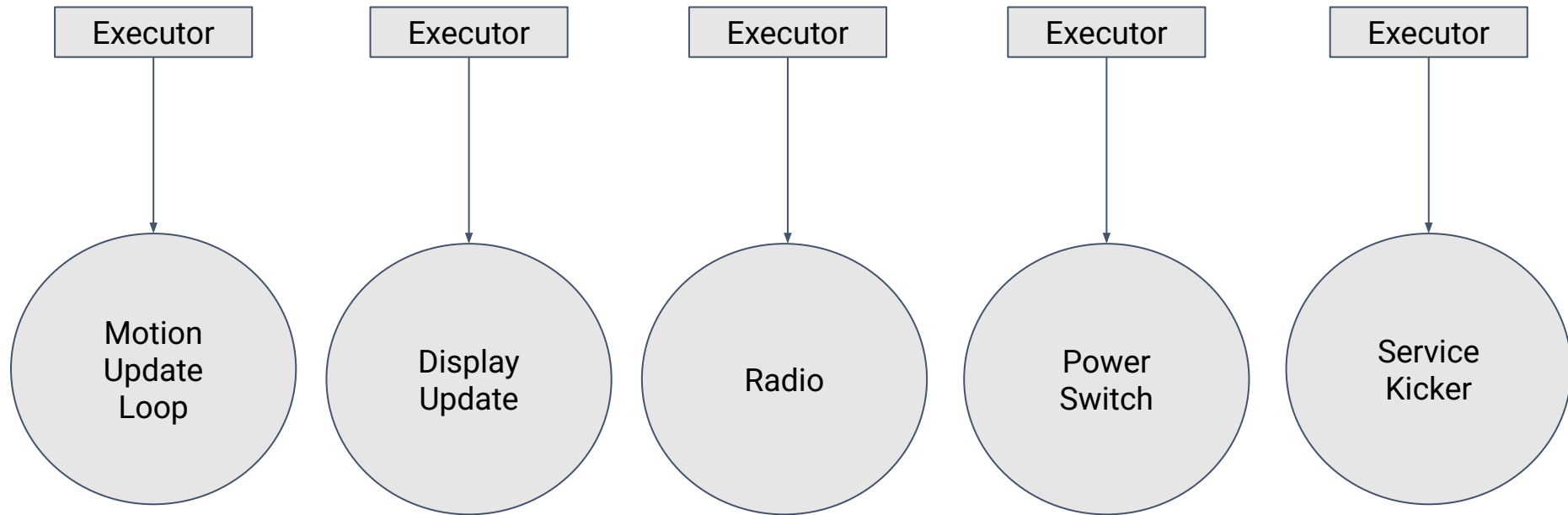
Motion
Update
Loop

Service
Kicker

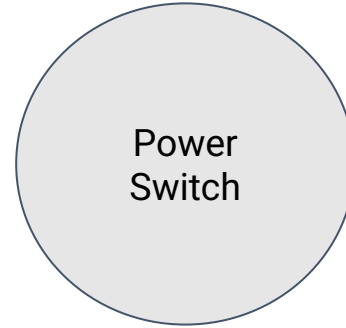
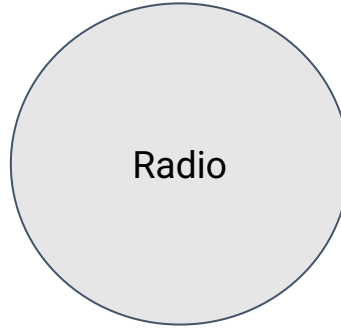
Radio

Display
Update

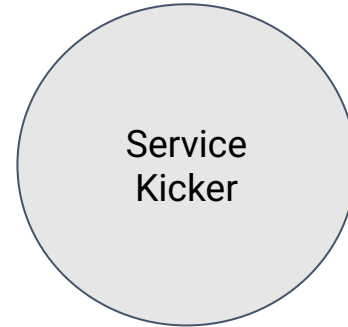
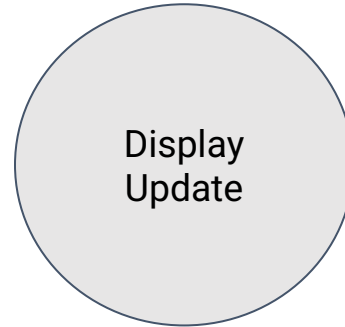
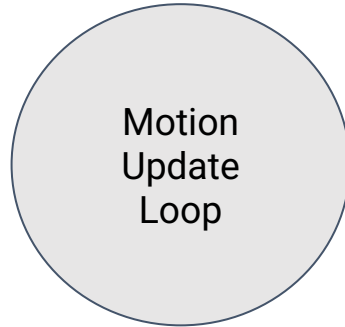
Power
Switch



Priority 1



Priority 2





Labs

State Machine Lock

